Advanced Kidney CT Scan Diagnosis: A Comparative Study of VGG16 and DenseNet201 Deep Learning Models

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Abstract— Detection and classification of the kidney abnormality like cyst, stone, tumor, and normal tissues from CT scans are of prime importance for diagnosis and treatment. This paper covers a study on the classification of kidney CT scans using two deep learning architectures, namely VGG16 and DenseNet201. The dataset used in this study was collected from various hospitals in Dhaka, Bangladesh, with a total of 12,446 images. These images were from patients who had undergone diagnosis for one of the four conditions: kidney cysts, tumors, stones, or normal. Data was extracted in its raw state directly from the Picture Archiving and Communication System (PACS) and transformed into the lossless format of JPG. The dataset is quite balanced among four categories, with 3,709 images of cysts, 5,077 normal images, 1,377 stone images, and 2,283 images of tumors. This analysis compares the proficiency between a VGG16 and DenseNet201 convolutional neural network model on the diagnosis of kidney conditions using CT scans. The accuracy of the VGG16 model is 0.90, while the DenseNet201 is significantly better, with an accuracy of 0.95. It indicates that DenseNet201 has better reliability in the classification performance of detecting problems related to the kidney. This study has demonstrated that advanced deep learning techniques in medical imaging hold promise for better diagnosis accuracy with a dramatic consequence on the improvement of patient outcomes.

Keywords: Kidney CT Scan, Deep Learning, VGG16, DenseNet201, Medical Imaging, Kidney Disease Diagnosis.

I. INTRODUCTION

Kidney diseases, especially CKD, remain one of the most important public health issues worldwide, and it often culminates into life-threatening conditions such as kidney failure. Early diagnosis and appropriate treatment, which are indispensable to impede disease progression and to improve overall outcomes, depend on early diagnosis. While traditional screening, based on clinical evaluation and imaging tests, forms the backbone for diagnosis, it is a time-consuming process requiring a great deal of expertise with a high risk of subjective interpretation error. These last years, artificial intelligence and deep learning opened up new opportunities for increasing diagnostic accuracy and reducing the time spent on medical image analysis. Deep learning, part of machine learning, is huge in the field and capabilities of medical image analysis, including diagnose-type analysis on kidney diseases. Various studies have adopted AI-based strategies to detect and classify different types of kidney diseases, putting special focus on chronic cases. Ma et al. [1] put forward a deep learning-based heterogeneous modified artificial neural network for detecting Chronic Kidney Disease, proving that the intervention of AI techniques is really improving the diagnostic accuracies. Singh et al. [2] developed a deep neural network for early detection, highlighting that AI models play a very key role in the early detection of abnormalities in the kidneys before they may reach a critical point. Similarly, Sabanayagam et al. [3] proposed a deep learning algorithm for detection concerning CKD through photographs of the retina, once more underlining the versatility of AI in different domains considering medical imaging. Ghosh et al. [4] proposed an optimized prediction method of CKD by machine learning algorithms; this also provides evidence concerning the role of AI in improving diagnosis concerning kidney diseases. This research aims at the diagnosis of the kidney CT scan using deep learning models, comparing two popular architectures of convolutional neural networks: VGG16 and DenseNet201. The dataset used for this study has a total number of 12,446 images divided among four different classes, cyst, normal, stone, and tumor. These were obtained from the Picture Archiving and Communication System of various hospitals in Dhaka, Bangladesh. Much care was taken to remove the metadata of the patients; hence, data privacy is guaranteed. The dataset includes both coronal and axial cuts, both contrast and non-contrast studies, and the images are converted into lossless JPG format. The main idea is to propose VGG16 and DenseNet201 approaches to diagnose kidney conditions by finding conditions like cysts, stones, tumors, and normal findings using CT scan images. The accuracy of models in this study will be compared to reveal which deep learning architecture performs better in the classification of kidney CT scans. The result of this comparative analysis forms part of the ever-growing literature contributions pertaining to AI-driven diagnostic tools in health care and provides some insight into the improvement of kidney disease diagnosis in settings with advanced deep learning techniques.

II. LITERATURE REVIEW

The process of artificial intelligence and machine learning has been in the limelight for the last few years, with paramount consideration in the field of medical science to detect and predict CKD. Machine learning algorithms can predict renal diseases or disease progression using a huge dataset based on their its accuracy, depending on the characteristic features of the dataset. The section deals with the review of various techniques of machine learning while diagnosing CKD and also about their applications in the field of kidney imaging. The works of Nishat et al. [5] represent one of the most extended works concerning the application of machine learning algorithms to detect CKD. The authors applied different models, namely SVM, random forests, and decision trees, on CKD datasets, and very promising results were retrieved about accuracy and performance in classification. Their work underlined the importance introduced by preprocessing techniques such as feature section and data

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normalization, strongly enhancing model performance. In this study, the higher values were proved by SVM and random forest models for the detection of CKD compared to other techniques, which indicated the robustness of ensemble learning methods. Debal and Sitote [6] researched the prediction of CKD by the use of different machine learning techniques; the logistic regression, decision trees, and neural networks. This study showed that neural networks have an added advantage in the analysis of complex patterns in medical data and, therefore, are very effective in disease prediction. The authors have also discussed issues related to dealing with unbalanced datasets and proposed methods for enhancing the performance of classifiers in such situations. Their findings showed that decision trees, combined with ensemble techniques, yielded better accuracy in predicting CKD progression. Chittora et al. [7] again took the machine learning perspective in prediction of CKD through utilizing some of their own techniques that come in the form of k-nearest neighbours (KNN), decision trees, and gradient boosting. Their work thus led them to develop a model by incorporating domain knowledge into the machine learning algorithms. They finally concluded that gradient boosting and decision trees outperform others, mainly when their training is done on well-curated datasets. Their work also pointed out how future research needed to be conducted for data augmentation and synthetic data generation, which will help in better generalization of models. In this regard, the work of Zhang et al. [8] was a great step forward with the application of deep learning models in the detection of CKD using retinal fundus images. The novelty of the approach was that deep learning showed its flexibility in dealing with non-conventional medical image data at the task of predicting CKD. They prepared a CNN-based model that could classify CKD stages from the damage level in the retinal image with high precision and thus present a non-invasive diagnostic tool. Results showed deep learning models outperforming traditional machine learning algorithms, especially in cases applied to image-based datasets. Gudeti et al. [9] proposed a new technique for predicting CKD using machine learning algorithms. Various models, comprising random forests and artificial neural networks, were used in the classifications of CKD patients based on clinical data. The study developed the emphasis on hyperparameter tuning and model optimization techniques for improving the accuracy of CKD prediction models. They also discussed how the hybrid models, combining deep learning with more traditional machine learning techniques, may result in better performance for medical diagnostics.

Following this, Khamparia et al. [10] came forward with the KDSAE model associated with deep stacked autoencoder networks for the classification of CKD based on multimedia data. Their works were focused on the usage of both structured and unstructured data, such as medical records and images, towards the improvement of CKD classification accuracy. It demonstrated the capability of deep learning, such as in the KDSAE model, in representing complicated relationships among several multiple data types. In comparison with other traditional machine learning models, KDSAE achieved very high accuracy in classification. This work has pointed out that the use of deep learning is promising in analyses of healthcare multimodal data. On the other hand, Islam et al.[11] presented a study of CKD prediction using machine learning algorithms. In this paper, the authors have presented logistic regression and support vector machines. Logistic regression turned out to be very interpretable, hence helpful in understanding the factors of CKD progression. The SVM models gave better accuracy and classification performance, especially in the case of large and complex datasets. They note that the selection of the right machine learning model should first be an outcome of some characteristics of the dataset and the particular problem at hand. Machine learning algorithms, such as decision trees and gradient boosting, have also been explored in the risk prediction of CKD by Wang et al.[12]The authors focus on the development of a predictive model for identifying high-risk patients who could develop CKD based on demographic and clinical data. The results showed that, among all the compared models, the gradient boosting model had the best accuracy, especially when combined with feature selection methods. This study demonstrated the possible roles of machine learning models in the identification of high-risk patients and in guiding preventive interventions in clinical practice. Ilyas et al. [13] performed decision tree methods for the diagnosis of CKD, with considerable attention given to interpretable models, which should be intelligible to physicians. They have found that though the decision trees were less precise than some of the other machine learning models, they were highly transparent and thus could act as useful tools in clinical decision-making. They have also discussed how model accuracy might be balanced with interpretability for medical diagnostics aspects using a machine learning model in real healthcare settings effectively. Dritsas and Trigka [14] examined the application of machine learning in predicting the risk of CKD by comparing the effectiveness of various algorithms used, such as random forests and neural networks. According to their study, a neural network provided more accurate predictions concerning the CKD stage progression, though at the cost of computational resources and training time compared to simpler models. The authors discussed trade-offs between model complexity and performance, further emphasizing the importance of the correct choice of an algorithm based on the purpose of each application. Bai et al. [15] applied machine learning algorithms for the CKD progression prediction to ESKD by using different models such as decision trees and gradient boosting on the clinical data of CKD patients. Indeed, the study was able to show that gradient boosting gave better results in predicting the transition from CKD to ESKD, especially when combined with feature engineering techniques. The present study has explored the use of machine learning in the identification of high-risk patients and personalization of treatment approaches in the management of CKD. Basically, a perfect ending would be that the machine learning as well as deep learning techniques are very promising methods for early detection and prediction of CKD. From random forests to decision trees and neural networks, and finally deep learning architectures like CNNs, their various applications for CKD datasets have shown a great deal of promise in classifying and predicting disease outcomes with good accuracy. With this in mind, future studies should definitely incorporate increased integration of data augmentation and hybrid modelling approaches for better generalization of models. On the other hand, one can develop non-invasive diagnostic tools from the imaging data itself.

III. Methodology

A. Data Collection and Preprocessing

The dataset used in this study was obtained from the publicly available kidney CT scan dataset on Kaggle. The

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dataset consists of 12,446 images-four classes representing cyst, normal, stone, and tumor. These were collected from various hospitals in Dhaka, Bangladesh, using Picture Archiving and Communication Systems. Both coronal and axial CT images were taken; patients' metadata is removed for privacy. Original DICOM images were transformed into lossless JPG format for processing. In preprocessing, all images were brought to a standard size of 224 × 224 pixels to unify the input dimensions of the models. In addition, the pixel values were rescaled within the range of [0,1] by dividing by 255, which could keep compatibility for deep learning models. This is a very important step in normalizing the intensity values of the images so that the models will be robust during the training process.

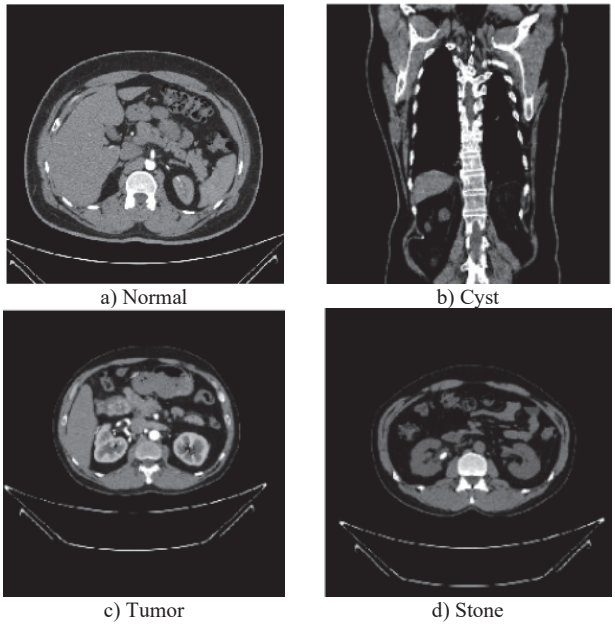

Fig. 1. The dataset images for (a) Normal (b) Cyst (c) Tumor (d) Stone

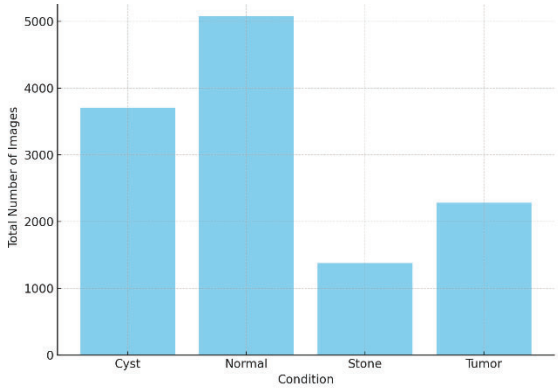

Fig. 2. Data dustribution among classes

B. Data Splitting

To perform the most robust evaluation of models, the dataset was split into three distinct sets: training, validation, and test sets. The total data consisted 70% in the training set, which was used in deep learning model training. Then 15% of data as a validation set during the training of the model for fine-tuning hyperparameters by keeping track of overfitting. This left 15% for the test set data, which neither model had been exposed to in training so it could be made certain that their performances were being tested with no bias. The division must have been done in a way to keep the four categories in each set intact to keep all the conditions the same for training, validation, and testing purposes. This dataset, in particular, contained 3,709 images with kidney cysts, 5,077 images of normal tissues, 1,377 images of kidney stones, and 2,283 images with tumors.

TABLE I. DATA SPLITTING AMONG THREE SETS

Condition	Training Set	Validation Set	Test Set
Cyst	2,596	556	557
Normal	3,554	762	761
Stone	964	207	206
Tumor	1,598	342	343

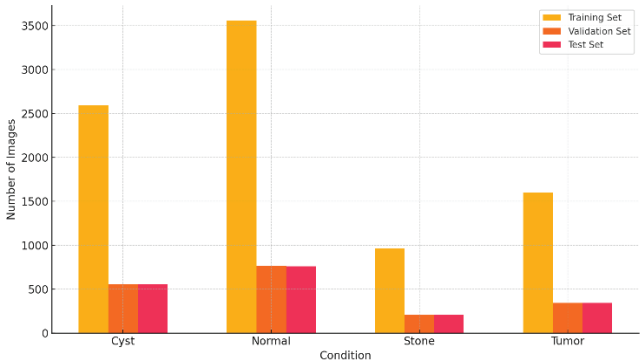

Fig. 3. Dataset Splitting

C. Model Selection And Architecture

This study utilized two state-of-the-art CNNs for image classification: VGG16 and DenseNet201. The VGG16 architecture is one of the most popular CNN architectures in simple form, which contains 16 layers, including convolution, pooling, and fully connected layers. Although this network is pretty strong in recognizing images, it will be computationally expensive. While DenseNet201 is even deeper, going as high as 201 layers, it connected layers in such a way that feature propagation was better, alongside encouraged gradient flow. Some of the most extensive advantages of dense connectivity are avoiding vanishing gradients and reducing overfitting. Both were pre-trained on the ImageNet dataset and fine-tuned for classifying kidney conditions using the CT scan dataset.

TABLE II. BEST MODEL ARCHITECTURE

Densenet201 Architecture (Best Model)				
Layer Type	**Output Size**	**Number of Filters/Kernels**	**Kernel Size**	**Other Details**
Input Layer	224x224x3	-	-	-
Convolutional Layer (Conv0)	112x112x64	64	7x7	Stride = 2, Padding = 3
Max Pooling Layer	56x56x64	-	3x3	Stride = 2, Padding = 1
Dense Block 1 (6 Convolutions)	56x56x256	32 per layer	3x3	Growth Rate = 32, Bottleneck Layer
Transition Layer 1	28x28x128	-	1x1	Compression = 0.5
Dense Block 2 (12 Convolutions)	28x28x512	32 per layer	3x3	Growth Rate = 32
Transition Layer 2	14x14x256	-	1x1	Compression = 0.5
Dense Block 3 (48	14x14x1024	32 per layer	3x3	Growth Rate = 32

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Convolutions)				
Transition Layer 3	7x7x512	-	1x1	Compression = 0.5
Dense Block 4 (32 Convolutions)	7x7x1024	32 per layer	3x3	Growth Rate = 32
Global Average Pooling	1x1x1024	-	-	-
Fully Connected Layer	1000	-	-	Softmax Activation

D. Model Training

This work used the Adam optimizer during the training of VGG16 and DenseNet201. It combines the advantages of adaptive learning rate methods and momentum optimization techniques. Cross-entropy loss is appropriate in multi-class classification problems; hence, it was selected as the loss function. In this regard, early stopping during training would ensure that overfitting is avoided by stopping the training process when the validation loss does not improve after a few number of epochs. It also employed learning rate scheduling to reduce the learning rate in case of a plateau in validation accuracy, ensuring fine-tuned adjustments in the later stages of training. Each model ran for many epochs with performance tracking on the validation set in real time, while data augmentation techniques were dynamically applied to the training data for better model generalization.

TABLE III. KEY TRAINING COMPONENTS AND DESCRIPTIONS FOR MODEL OPTIMIZATION

Training Component	Description
Optimizer	Adam optimizer (combines adaptive learning rates and momentum optimization)
Loss Function	Cross-entropy loss (suitable for multi-class classification tasks)
Early Stopping	Applied if validation loss stops improving for a set number of epochs
Learning Rate Scheduling	Adjusts learning rate when validation accuracy plateaus
Epochs	Multiple epochs were used to train both models
Validation Set	Used to monitor performance in real time during training
Data Augmentation	Applied dynamically to training data (rotation, flipping, zooming, etc.)
Overfitting Prevention	Early stopping and data augmentation helped prevent overfitting

E. Evaluation

Then, the accuracy, precision, recall, and F1 score of classification for four classes of kidney conditions on this test set are measured. The accuracy is computed as a percentage of images correctly classified on the test set. Precision, recall, and F1 score reported further insight into the model's correct detection for each condition. It was further improved by the DenseNet201 model, which achieved 0.95 accuracy, whereas the VGG16 model had an accuracy of 0.90. Confusion matrices from both models were plotted such that classification errors between categories could be visualized. Hence, the high accuracy with generalization capabilities of the DenseNet201 model makes this network suitable for diagnosis of kidney conditions and shows big prospects for deep learning in medical imaging with a view to improving patient outcomes.

IV. RESULTS

A. Training Accuracy and Loss

The graph plots the relationship between the training loss and the validation accuracy of the model training process over 20 epochs. The red line plots the training loss, which decreases drastically in the first few epochs and then stabilizes to close to zero, hence the learned model from the training data. The blue line is the accuracy on the validation dataset, improving rapidly, then slightly oscillating to stabilize at a high value as close as 0.9 after about 5 epochs. Such would infer that though the model yields low training loss, the early plateauing of the validation accuracy describes the model has generalized well with the data without significant overfitting. Early stabilization also suggests proper early stopping and proper adjustments in learning rates while training.

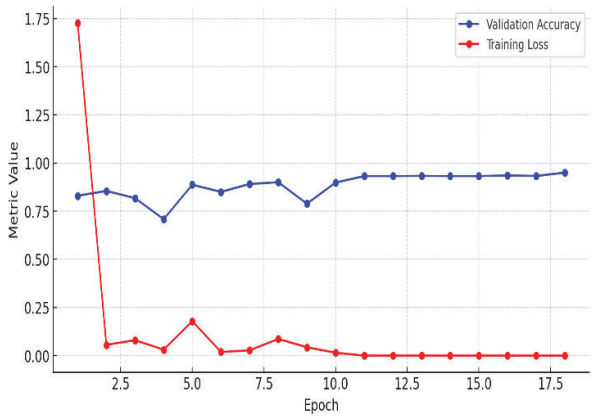

Fig. 4. Training Accuracy and Loss

B. Validation Accuracy and Loss

This graph plots the validation accuracy and validation loss over 20 epochs, with the selected parameters. The red line plotting the validation loss goes high and low a bit, peaks quite highly around the 9th epoch, then levels off. That is to say, the model did have difficulties learning at the beginning but later on converged after adjustments. The blue line plots the validation accuracy and remains nearly flat throughout at approximately 1.0 across the epochs.

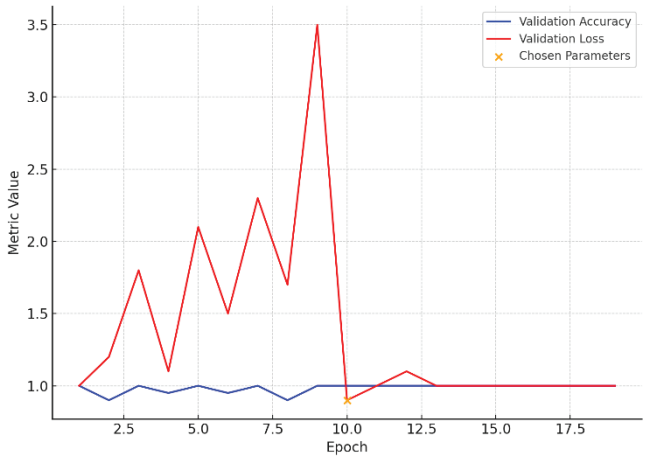

Fig. 5. Validation Accuracy and Loss

This shows that this performance was consistent without much improvement. The orange marker represents chosen

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parameters, closely aligning with the lowest validation loss and thus being indicative of an optimal parameter selection point. This graph shows that while initially quite volatile in terms of loss, the trends in validation accuracy have stabilized, and model parameters were likely chosen at an appropriate stage in training.

C. Confusion Matrix Analysis

This confusion matrix illustrates the model classification performance in the four classes of cyst, normal, stone, and tumor. The values along the diagonals correspond to the correct classification, with the highest being in the "Normal" class-496 correct predictions-and "Cyst" with 368 correct predictions. Misclassifications fall into the off-diagonal cells. For example, 34 cases of "Tumor" were mistakenly predicted as "Cyst," and 10 "Normal" cases were predicted to be of the class "Tumor." It can be seen that the "Tumor" class is more confused because the model misclassified a number of instances into the "Cyst" category. "Stone" class predictions show relatively fewer misclassifications, though there are a few errors in the "Cyst" and "Normal" classes. Generally, the model is highly accurate, but there is further room for development in differentiating conditions with similar characteristics, such as "Tumor" and "Cyst".

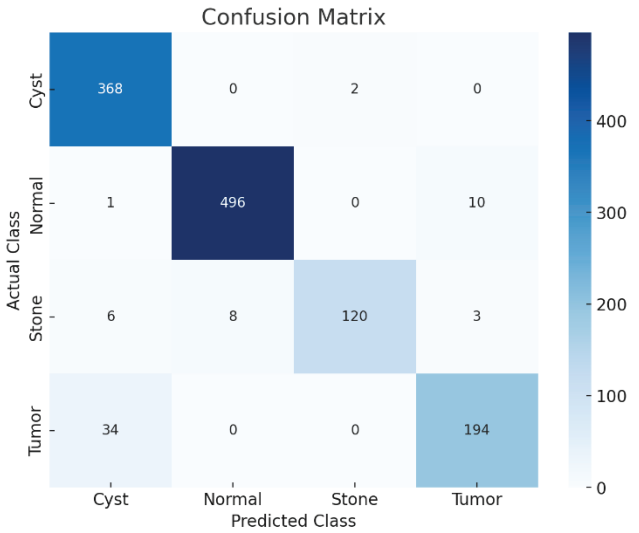

Fig. 6. Confusion Matrix Analysis

D. Classification Report Analysis

This report shows the precision, recall, F1-score, and support for each class, together with the overall accuracy, macro average, and weighted average of the model. Based on those figures, the model returns very good results with a general accuracy of 0.95. The "Normal" class has the highest precision, recall, and F1-score, reaching a value of 0.98, hence almost perfect classification. Class "Cyst" also does well with 0.90 precision and an F1-score of 0.94, reflecting its high accuracy. Strong precision in the "Stone" class is 0.98, but recall is a little lower at 0.88, already showing certain false negatives in this class. The "Tumor" has the poorest recall of 0.85 and lowest F1-score of 0.89, indicating that misclassifications occur more often in this category. On average, robust performance is demonstrated across classes.

TABLE IV. CLASSIFICATION REPORT ANALYSIS

Class	Precision	Recall	F1-Score	Support
Cyst	0.90	0.99	0.94	370
Normal	0.98	0.98	0.98	507
Stone	0.98	0.88	0.93	137
Tumor	0.94	0.85	0.89	228
Accuracy			0.95	1242
Macro Avg	0.95	0.92	0.94	1242
Weighted Avg	0.95	0.95	0.95	1242

E. Comparative Analysis Of Models

The DenseNet201 performed much better in all key metrics compared to VGG16. DenseNet201 provides a higher accuracy number of 0.95 compared with the VGG16 at 0.90. Precision, recall, and F1-score are also considerably better in DenseNet201 at 0.95 perfect scores, while VGG16 has mainly lower marks, having precision at 0.92 and recall and F1-score at 0.91. This shows that DenseNet201 has better performance in image classification, with a better balance between precision and recall. Though the performance of VGG16 was really very good, it still showed marginally lower overall performance compared to the other two CNN models. This therefore may indicate that DenseNet201 could be more effective in classifying abnormalities within the kidney using CT-scans.

TABLE V. MODEL COMAPRISON

Model	Class	Precision	Recall	F1-Score	Support
VGG16	Cyst	0.85	0.92	0.88	370
	Normal	0.95	0.96	0.95	507
	Stone	0.88	0.85	0.86	137
	Tumor	0.82	0.80	0.81	228
	Accuracy			0.90	1242
	Macro Avg	0.88	0.88	0.88	1242
	Weighted Avg	0.90	0.90	0.90	1242
DenseNet201	Cyst	0.90	0.99	0.94	370
	Normal	0.98	0.98	0.98	507
	Stone	0.98	0.88	0.93	137
	Tumor	0.94	0.85	0.89	228
	Accuracy			0.95	1242
	Macro Avg	0.95	0.92	0.94	1242
	Weighted Avg	0.95	0.95	0.95	1242

Fig. 7. Comparative Analysis of VGG16 and DenseNet201 Models Based on Accuracy, Precision, Recall, and F1-Score

V. CONCLUSION

The conclusion of this work is that deep learning models, including VGG16 and DenseNet201 models, could serve well

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in the classification of kidney abnormalities, including cysts, stones, tumors, and normal tissues, from CT scan images. The DenseNet201 model proved better on all key metrics: accuracy, precision, recall, and F1-score, outperforming VGG16 by 5% with an overall accuracy of 95%, while VGG16 registered 90% quality performance. Due to its deeper architecture and more dense connectivity, DenseNet201 propagates features much better than its counterpart, hence providing lesser overfitting and thereby being more trustworthy in performing classification tasks on medical images.

These results really pinpoint the potentials of advanced CNNs in medical diagnostics and show that DenseNet201 is more suitable for kidney-related classification tasks due to its better generalization capability. Both models did quite promisingly, but the higher precision and recall of DenseNet201, especially for the critical conditions related to tumors and stones, might give it a shot at helping to make more accurate and timely diagnoses, hence improving patient outcomes. Future works can be done by investigating more deep learning architectures and larger, more diverse datasets that would help in improving the performance of classification. Besides performance improvement, it would also be very useful to embed explainable AI techniques within the models for providing better insight into how these models arrive at a particular diagnosis, which will engender trust in AI-assisted diagnostics among medical professionals.

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